

SIMATS Engineering

SIMATS Online Programming Lab — Python Programming (Class Test)

Course Name	Python Programming
Course Code	CSA0804
Name	Monish.L
Register Number	1972260035
Topic	CO3-AT2-CLASS TEST
Question	Q5 — List Comprehensions on a Number List (1 Mark)

Q5. [1 Mark]

Given nums = [1,2,3,4,5,6,7,8,9,10], write list comprehensions to create:

- (a) Cubes of all numbers.
 - (b) Only odd numbers.
 - (c) Numbers divisible by 3.
- Display all three results.

Approach / Design

- **Cubes:** [n ** 3 for n in nums] applies the cube operation to every element of nums in a single comprehension.
- **Odd numbers:** [n for n in nums if n % 2 != 0] filters the list, keeping only elements whose remainder on division by 2 is non-zero.
- **Divisible by 3:** [n for n in nums if n % 3 == 0] filters the list, keeping only elements that are exact multiples of 3.
- **Display:** each resulting list is printed with a descriptive label.

Python Program

```
"""
CO3-AT2 - Class Test
Q5: List Comprehensions on a Number List [1 Mark]
-----
Given nums = [1,2,3,4,5,6,7,8,9,10], uses list comprehensions to create:
(a) Cubes of all numbers.
(b) Only odd numbers.
(c) Numbers divisible by 3.
Displays all three results.
"""

nums = [1, 2, 3, 4, 5, 6, 7, 8, 9, 10]

# (a) Cubes of all numbers
cubes = [n ** 3 for n in nums]

# (b) Only odd numbers
odd_numbers = [n for n in nums if n % 2 != 0]

# (c) Numbers divisible by 3
divisible_by_3 = [n for n in nums if n % 3 == 0]
print("Cubes:", cubes)
```

```
print("Odd Numbers:", odd_numbers)
print("Divisible by 3:", divisible_by_3)
```

Sample Output

```
Cubes: [1, 8, 27, 64, 125, 216, 343, 512, 729, 1000]
Odd Numbers: [1, 3, 5, 7, 9]
Divisible by 3: [3, 6, 9]
```

Conclusion

Each requirement maps cleanly onto a single list comprehension: a transformation expression for the cubes, and a filtering condition for the odd numbers and the multiples of 3. Running the program on `nums = [1..10]` correctly produces the cubes `[1, 8, 27, ..., 1000]`, the odd numbers `[1, 3, 5, 7, 9]`, and the multiples of 3 `[3, 6, 9]`, confirming that all three comprehensions behave as expected.